What holds a paper boat?

01 Can the shape of paper help it carry a penny?



An imagined scene. Penny counts and time in the lessons are pretend.

Look closer. Compare the flat sheet with the folded boat in the next scene.

A little explanation. Folded walls and a broad, flat bottom make a space that keeps water out. The air inside is part of what floats. A flat sheet may rest on water briefly, but it has no walls to protect that space when a load pushes it down.

Try this. Imagine folding the sides up. Which part would keep water out?

02 What is holding the boat up?



An imagined scene. Penny counts and time in the lessons are pretend.

Look closer. Look where the water meets the boat. The bottom of the tub is far below.

A little explanation. The boat displaces water: it pushes water aside. Water pushes up with a force as big as the weight of the water pushed aside. The boat floats when that upward push equals the boat's weight.

Try this. Use your voice to follow the steps: push water aside, feel water push up, float.

03 Why does the waterline climb?



An imagined scene. Penny counts and time in the lessons are pretend.

Look closer. The pennies are inside. The water reaches higher on the sides.

A little explanation. Pennies add load, the weight the boat carries. To balance that extra weight, the boat sinks lower and pushes aside more water. Too much load, or a tip, lets water spill over the sides and swamp the boat.

Try this. Predict what would happen to the waterline if Little removed a penny.

04 Why can't a paper boat float forever?



An imagined scene. Penny counts and time in the lessons are pretend.

Look closer. The damp paper has sagged. Its crisp folds are softer.

A little explanation. Paper is made of tiny plant fibers that absorb water. Wet paper gets heavier and softer and loses its shape. A crayon or wax coat slows the soaking, but it does not make the boat last forever.

Try this. Choose one command for another voyage, such as "Watch the waterline."